



FlowRoute

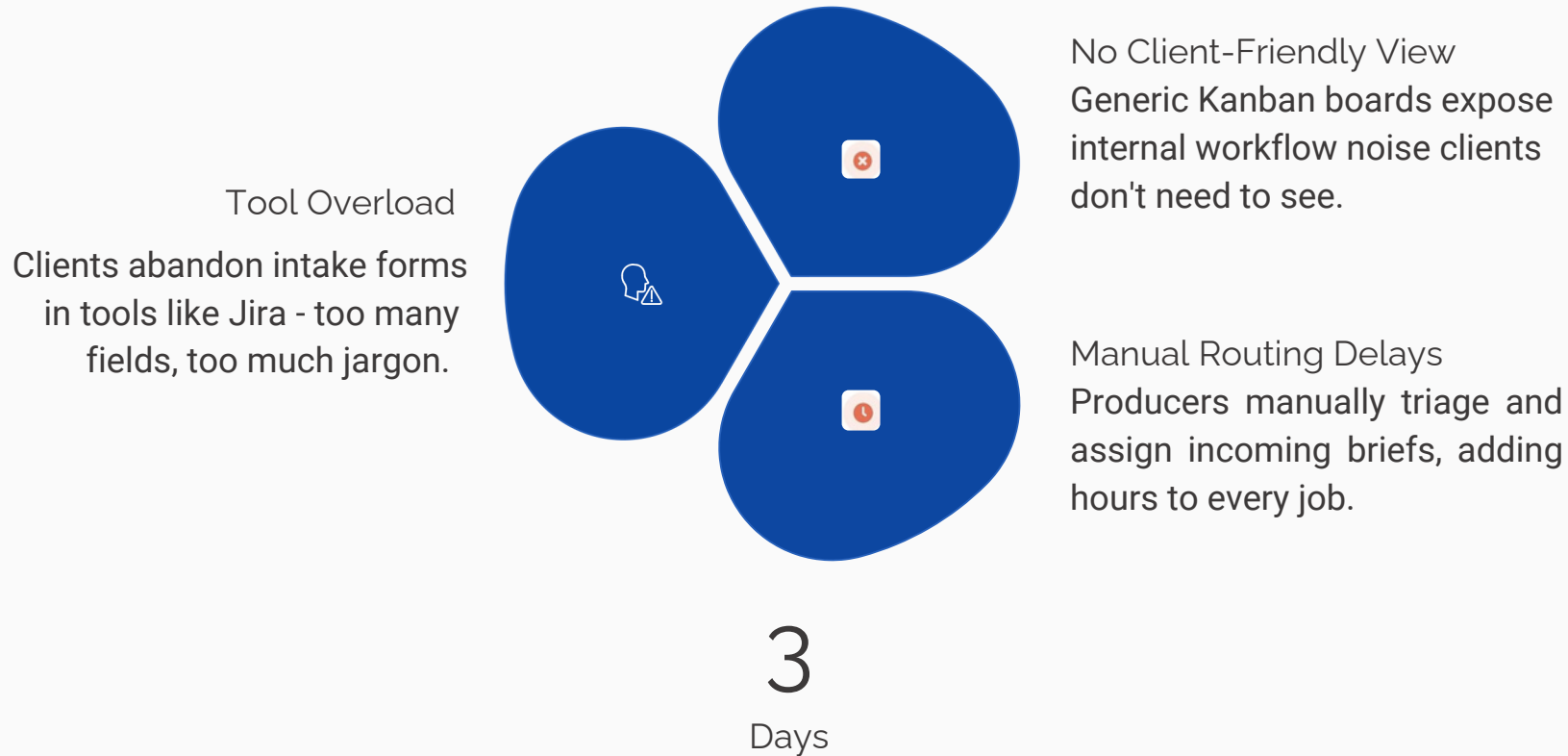
Automated Workflow & Resource Manager

Design Presentation - System Requirements & Software Design Specification

Team: FlowRoute | Supervisor: Dr. Nalin Warnajith | Date: June 2026

The Problem

Videoproductionagenciesareforcedinto project tools built for software teams



average manual brief-to-assignment turnaround reported by our client

"We just need clients to drop a file and a brief - not learn an entire PM system."

- Client Interview #2, Sept 2025

Who We're Building For

Three actor groups defined in the SRS use-case model

Client

Submits video briefs and tracks delivery status. Wants minimal friction.

Producer

Triages incoming requests and assigns them to the right editor or team.

Routing Engine

Automated actor (Python service) that scores and assigns jobs by priority and load.

SRS Summary - Core Requirements

Five functional requirement clusters derived from client interviews and competitive analysis



Client Intake

FR-01-03: Drag-and-drop upload, project metadata form, success/failure states



Auto-Routing

FR-04-06: Priority-based assignment of briefs to producers/editors



Status Tracking

FR-07-09: Real-time project status visible to client and team



Delivery

FR-10-12: Final asset packaging and secure client download

System Architecture

Layered architecture-decided in ADR-01, optimized for independent scaling of the routing engine



Client Tier

React SPA



Gateway Tier

API Gateway



Application Tier

Node.js API + Python Routing Engine



Data Tier

MongoDB + File Storage

Why This Stack - ADR-01

Decision

Node.js API + separate Python service for routing logic

Rationale

- Routing/assignment logic benefits from Python's data/ML ecosystem for future scoring improvements
- Decouples scaling - routing engine can scale independently under load spikes
- Node.js keeps the REST/auth layer in the team's strongest skill set

Alternatives Considered

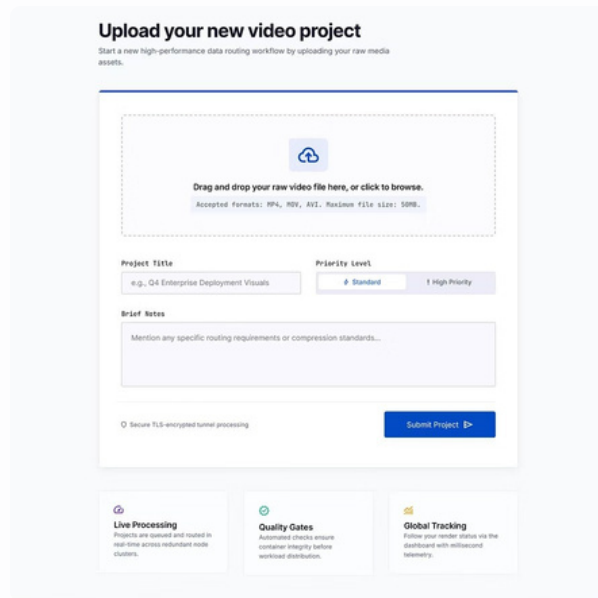
Option	Trade-off	Verdict
Monolith (Node only)	Simpler, but couples routing logic to API scaling	Rejected
Full microservices	Over-engineered for an 8-week design scope	Rejected
Node + Python (chosen)	Slight ops overhead, but clean separation of concerns	Accepted



Quality attributes addressed: scalability, maintainability, independent deplorability

UI Prototype - Client Intake Portal

Three-screen flow: minimal, drag-and-drop first, clear success/failure states



Upload your new video project
Start a new high-performance data routing workflow by uploading your raw media assets.

Drag and drop your raw video file here, or click to browse.
Accepted formats: MP4, MOV, AVI. Maximum file size: 50MB.

Project Title
e.g., Q4 Enterprise Deployment Visuals

Priority Level
Standard High Priority

Brief Notes
Mention any specific routing requirements or compression standards...

Secure TLS-encrypted tunnel processing

Submit Project

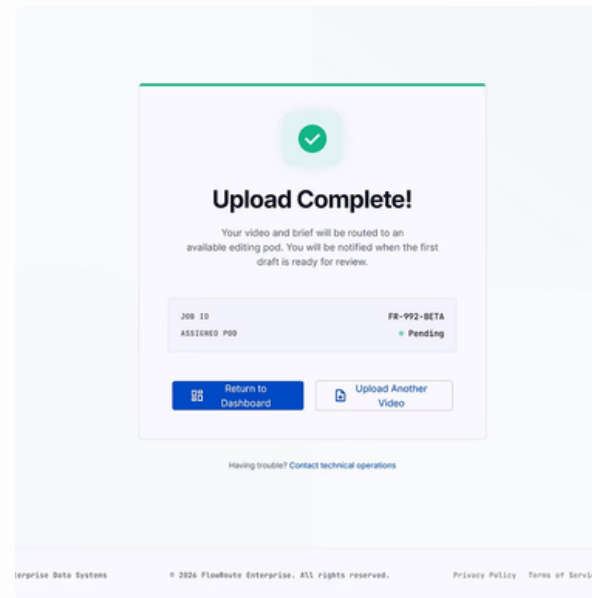
Live Processing
Projects are queued and routed in real-time across redundant node clusters.

Quality Gates
Automated checks ensure container integrity before workload distribution.

Global Tracking
Follow your media status via the dashboard with millisecond telemetry.

01 - Upload Form

Drag-and-drop zone + project title, priority, and brief fields



Upload Complete!
Your video and brief will be routed to an available editing pod. You will be notified when the first draft is ready for review.

JOB ID
ASSIGNED POD

FR-992-BETA
Pending

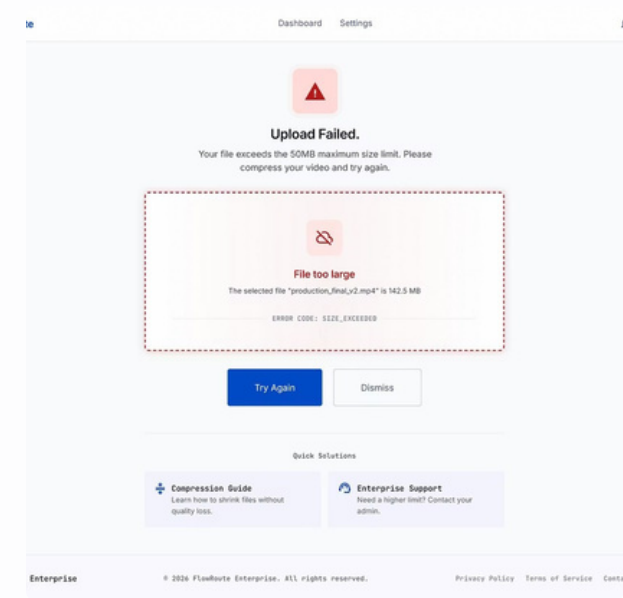
Return to Dashboard Upload Another Video

Having trouble? [Contact technical operations](#)

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02 - Success State

Confirmation with reference ID and submission summary



Dashboard Settings

Upload Failed.
Your file exceeds the 50MB maximum size limit. Please compress your video and try again.

File too large
The selected file "production_final_v2.mp4" is 142.5 MB
ERROR CODE: SIZE_EXCEEDED

Try Again Dismiss

Quick Solutions

Compression Guide
Learn how to shrink files without quality loss.

Enterprise Support
Need a higher limit? Contact your admin.

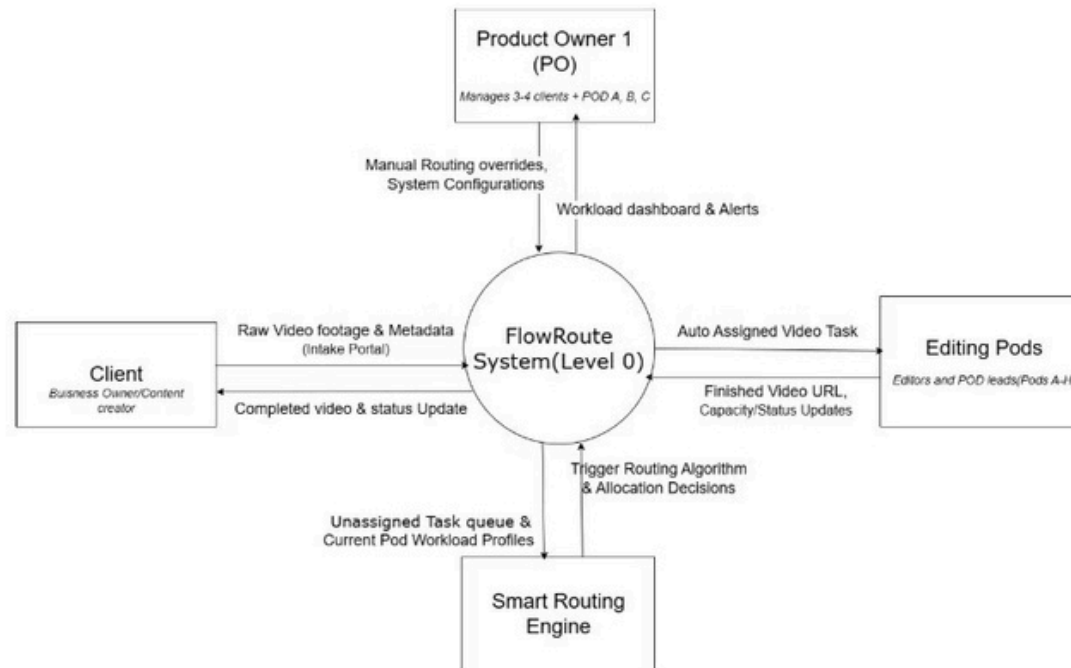
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03 - Failure State

Inline, specific error messages with retry path

From Design to Development

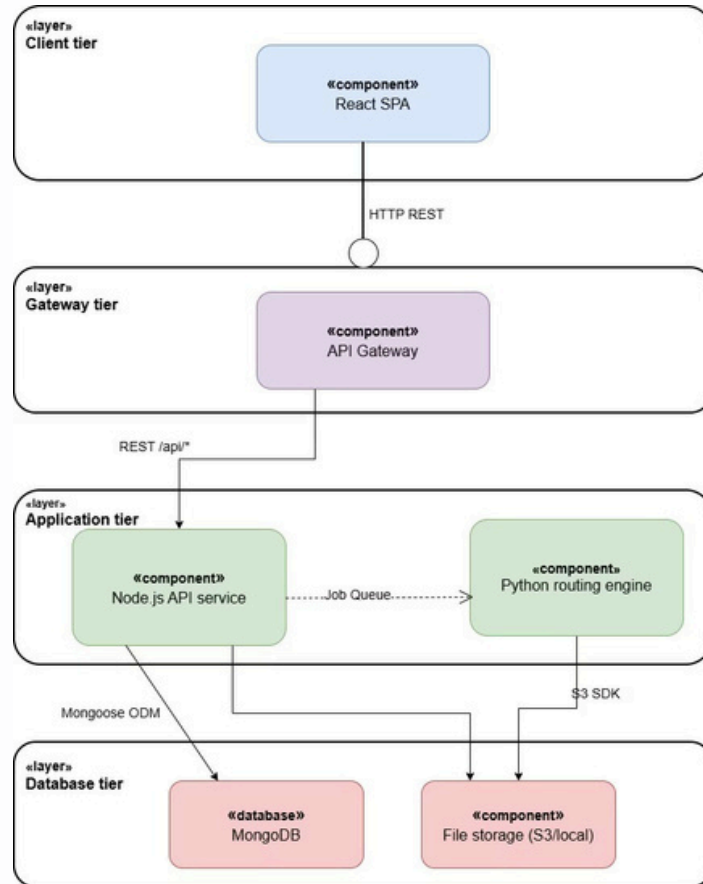
What's locked in for SENG 34213, and what's still open



Finalized

- SRS with 12 functional requirements, signed off
- Architecture (ADR-01) and component diagrams
- Client intake portal wireframes, team-approved
- Database schema for projects, users, assets

From Specification to Implementation



- Routing algorithm scoring weights - needs real usage data
- Authentication provider selection (Auth0 vs. custom JWT)
- Load testing plan for concurrent uploads
- CI/CD pipeline - GitHub Actions becomes mandator



Thank You

Questions & Discussion